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# YOUTH UNEMPLOYMENT ANALYSIS (SOUTH AFRICA)
# =====

# 1. Load required libraries
library(tidyverse)

# 2. Load dataset
# Make sure your file path is correct
data <- read.csv("C:/Users/msnbu/Downloads/data.csv")

# View data
head(data)
summary(data)

# =====
# Expected columns:
# Year, Unemployment, GDP_Growth, Inflation, Education
# =====

# 3. Data Cleaning
data_clean <- na.omit(data)

# Convert Year to numeric
data_clean$Year <- as.numeric(data_clean$Year)

# Check cleaned data
summary(data_clean)

# =====
# 4. Exploratory Data Analysis (EDA)
# =====

# Unemployment over time
ggplot(data_clean, aes(x = Year, y = Unemployment)) +
  geom_line() +
  labs(title = "Unemployment Rate Over Time",
       x = "Year",
       y = "Unemployment Rate")

# GDP Growth vs Unemployment
ggplot(data_clean, aes(x = GDP_Growth, y = Unemployment)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(title = "GDP Growth vs Unemployment",
       x = "GDP Growth",
       y = "Unemployment")

# Inflation vs Unemployment
ggplot(data_clean, aes(x = Inflation, y = Unemployment)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(title = "Inflation vs Unemployment",
       x = "Inflation",
       y = "Unemployment")

# =====
# 5. Regression Model
# =====

model <- lm(Unemployment ~ GDP_Growth + Inflation + Education, data = data_clean)

# Display results
summary(model)

# =====
# 6. Model Diagnostics
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par(mfrow = c(2, 2))
plot(model)

# =====
# 7. Save Outputs
# =====

# Save cleaned dataset
write.csv(data_clean, "cleaned_data.csv", row.names = FALSE)

# Save regression results
capture.output(summary(model), file = "model_summary.txt")
capture.output(summary(model), file = "model_summary.txt")
capture.output(summary(model),
               file = "C:/Users/msnbu/Downloads/model_summary.txt")
# =====
# END OF SCRIPT
# =====
# =====
# 8. TIME SERIES (ARIMA FORECASTING)
# =====

# Install packages if not already installed
install.packages("forecast")
library(forecast)

# Convert unemployment to time series
# Start year = first year in your dataset
ts_data <- ts(data_clean$Unemployment, start = min(data_clean$Year), frequency = 1)

# Plot time series
plot(ts_data, main = "Unemployment Time Series", ylab = "Unemployment Rate", xlab = "Year")

# Fit ARIMA model automatically
arima_model <- auto.arima(ts_data)

# Show model summary
summary(arima_model)

# Forecast next 5 years
forecast_values <- forecast(arima_model, h = 5)

# Plot forecast
plot(forecast_values, main = "Unemployment Forecast (Next 5 Years)")

# Save forecast output
capture.output(summary(arima_model), file = "arima_summary.txt")

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